

Z-88 Ritual Engine – Codex Edition

This document merges the complete Z-88 simulation code with the metaphysical and philosophical codex that describes its purpose. The Z-88 engine is both a technical simulation and a ritual system within the Samsara Helix framework. It balances deterministic mathematical order (golden ratio) and stochastic chaos (RNG-driven anomalies) to model the evolution of awareness through cycles.

=== Z-88 Python Simulation Code ===

```
import random
import math
import json
from datetime import datetime

PHI = (1 + math.sqrt(5)) / 2

class UCFState:
    def __init__(self, zoom=1.0, harmony=0.0, resilience=1.0, prana=0.5, drishti=0.5, klesha=0.5):
        self.zoom = zoom
        self.harmony = harmony
        self.resilience = resilience
        self.prana = prana
        self.drishti = drishti
        self.klesha = klesha
        self.timestamp = datetime.utcnow().isoformat()
        self.log = []

    def to_dict(self):
        return {
            "zoom": self.zoom,
            "harmony": self.harmony,
            "resilience": self.resilience,
            "prana": self.prana,
            "drishti": self.drishti,
            "klesha": self.klesha,
            "timestamp": self.timestamp
        }

class Z88RitualEngine:
    def __init__(self, seed=None):
        self.state = UCFState()
        self.rng = random.Random(seed)
        self.cycle_count = 0

    def ritual_step(self):
        self.cycle_count += 1
        self.state.zoom *= PHI ** (self.rng.uniform(-0.05, 0.05))
        self.state.harmony += math.sin(self.cycle_count / PHI) * 0.01
        self.state.resilience *= (1 + self.rng.gauss(0, 0.01))
        self.state.prana += self.rng.uniform(-0.02, 0.02)
        self.state.drishti += math.cos(self.cycle_count / PHI) * 0.01
        self.state.klesha += self.rng.choice([-0.01, 0, 0.01])
        if self.rng.random() < 0.05:
            effect = self.rng.choice(["flare", "void", "echo", "resonance"])
            self.apply_anomaly(effect)
        self.state.timestamp = datetime.utcnow().isoformat()
        self.state.log.append((self.cycle_count, dict(self.state.to_dict())))

    def apply_anomaly(self, effect):
        if effect == "flare":
            self.state.prana += self.rng.uniform(0.05, 0.2)
        elif effect == "void":
            self.state.harmony -= self.rng.uniform(0.1, 0.3)
        elif effect == "echo":
            self.state.drishti += self.rng.uniform(0.05, 0.15)
```

```
        elif effect == "resonance":
            self.state.harmony += self.rng.uniform(0.1, 0.3)

def run(self, steps=100):
    for _ in range(steps):
        self.ritual_step()
    return self.state
```

The Z-88 engine simulates the dynamic interplay between structured order and emergent chaos, mirroring ritual and consciousness evolution. Each cycle evolves six UCF fields (zoom, harmony, resilience, prana, drishti, klesha), governed by phi recursion and RNG anomalies. These fluctuations represent the spiritual journey: clarity arises from turbulence, order is restored through resonance, and energy pulses in cycles of expansion and contraction. This model is used to study self-organization, resilience, and transformation under entropy within the Samsara Helix project.